Microsoft Elevate

CAPSTONE PROJECT

HOTEL INSIGHT DASHBOARD

PRESENTED BY

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OUTLINE:

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- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
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PROBLEM STATEMENT:

The hospitality industry generates large volumes of operational data such as revenue, occupancy rate, average daily rate (ADR), RevPAR, and customer ratings across multiple hotels and cities. However, this data is often scattered across different systems, making it difficult for hotel management to quickly analyze performance and make informed business decisions.

The challenge is to design an interactive analytics dashboard that consolidates hotel performance data and provides clear insights into key metrics such as revenue trends, occupancy rates, property performance, and customer ratings. This dashboard should enable hotel managers to monitor performance across properties, identify high-performing hotels, detect operational inefficiencies, and support data-driven decision-making to improve profitability and customer satisfaction.

PROPOSED SOLUTION:

- To address the challenge of scattered hotel performance data and difficulty in analysis, an interactive **Hotel Insight Dashboard** is developed using **data visualization and business intelligence tools**. The dashboard integrates hotel data from multiple properties and presents it in a centralized, easy-to-understand format.

- The solution provides visual insights into key performance indicators such as: **Revenue, RevPAR, ADR, Occupancy Rate, Realisation Percentage, and Average Ratings**.

- Interactive filters for **city, room type, class, and time period** allow users to analyze performance dynamically across different hotels and locations.

- Charts and visualizations such as **bar charts, line graphs, and donut charts** help management quickly compare hotel performance, track revenue distribution, monitor customer satisfaction, and identify trends over time.

This dashboard enables hotel managers and stakeholders to **make faster data-driven decisions, optimize operational strategies, improve occupancy rates, and enhance overall business performance**.

SYSTEM APPROACH:

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the dashboard.

•**System requirements:** Power BI Desktop, Windows Operating System.
•**Technology Used:** Microsoft Power BI for visualization, DAX (Data Analysis Expressions) for calculated measures, and Power Query for data transformation.

Finally, **interactive dashboard visualizations** are created using charts, graphs, KPI cards, and filters. These visual elements allow users to analyze hotel performance by **city, property, category, and time period**, making it easier for management to monitor trends and support strategic decision-making.

ALGORITHM & DEPLOYMENT:

Algorithm

1.Start
2.Collect hotel dataset containing information such as property details, revenue, occupancy, ADR, RevPAR, ratings, and booking data.
3.Perform data preprocessing:
- •Remove duplicate records
- •Handle missing values
- •Format columns and data types correctly

4.Import the cleaned dataset into the Business Intelligence tool.
5.Create relationships between tables to build the data model.
6.Calculate key metrics such as:
- •Revenue
- •RevPAR (Revenue per Available Room)
- •ADR (Average Daily Rate)
- •Occupancy Percentage
- •Realisation Percentage

7.Design visualizations such as KPI cards, bar charts, line charts, and donut charts to represent the metrics.
8.Add interactive filters for city, room type, hotel class, and time period.
9.Test the dashboard to ensure correct calculations and interactive functionality.
10.Display insights to help hotel management analyze performance and make decisions.
11.End.

ALGORITHM & DEPLOYMENT:

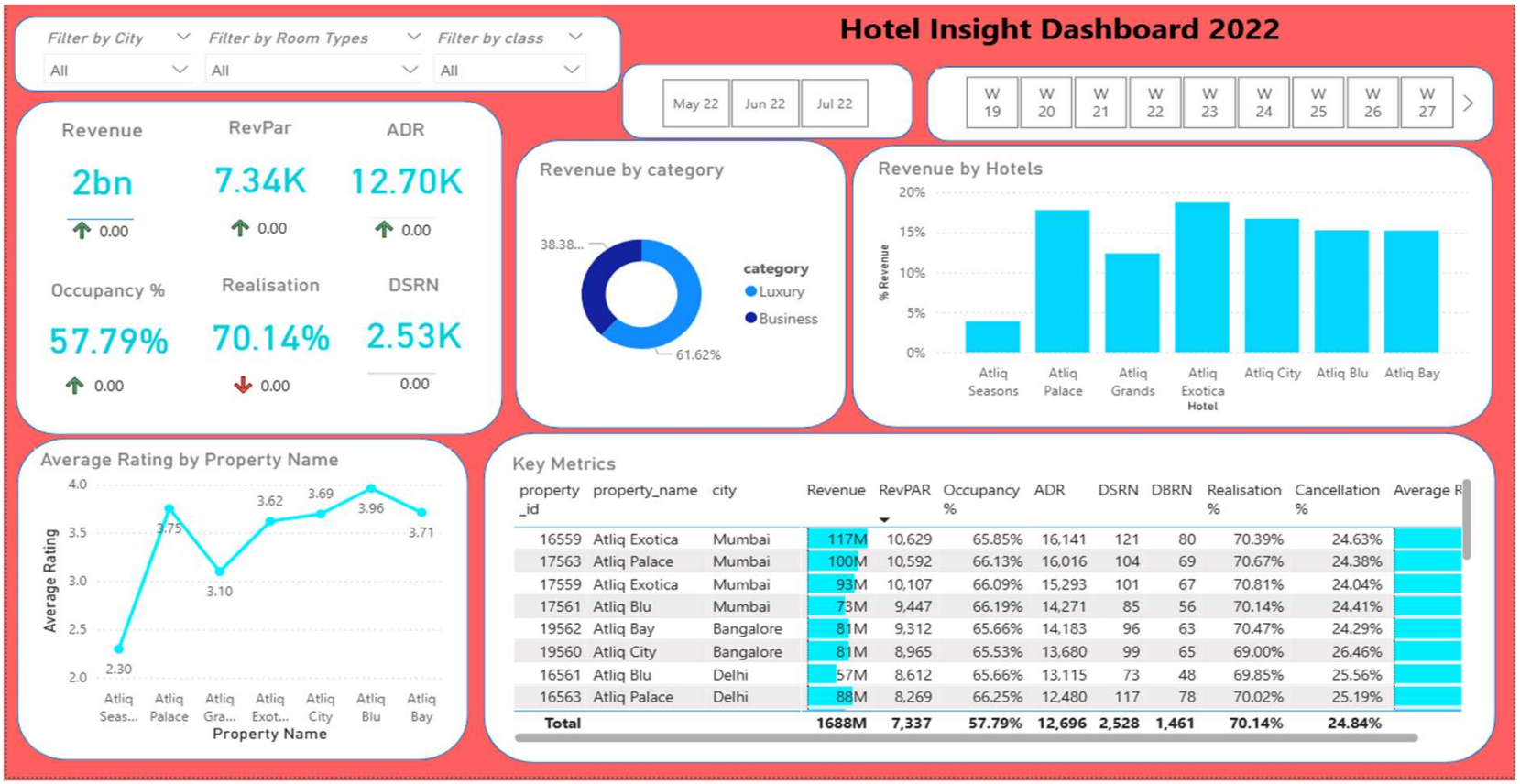
Deployment

The Hotel Insight Dashboard is deployed using a Business Intelligence platform to make it accessible for users and decision-makers.

1.The dashboard is developed using a data visualization tool such as **Power BI**.
2.After development, the dashboard is published to the **Power BI Service or BI server**.
3.Users such as hotel managers and analysts can access the dashboard through a web browser.
4.The system supports real-time or periodic data updates to keep insights current.
5.Security and access permissions are applied to ensure only authorized users can view or edit the dashboard.

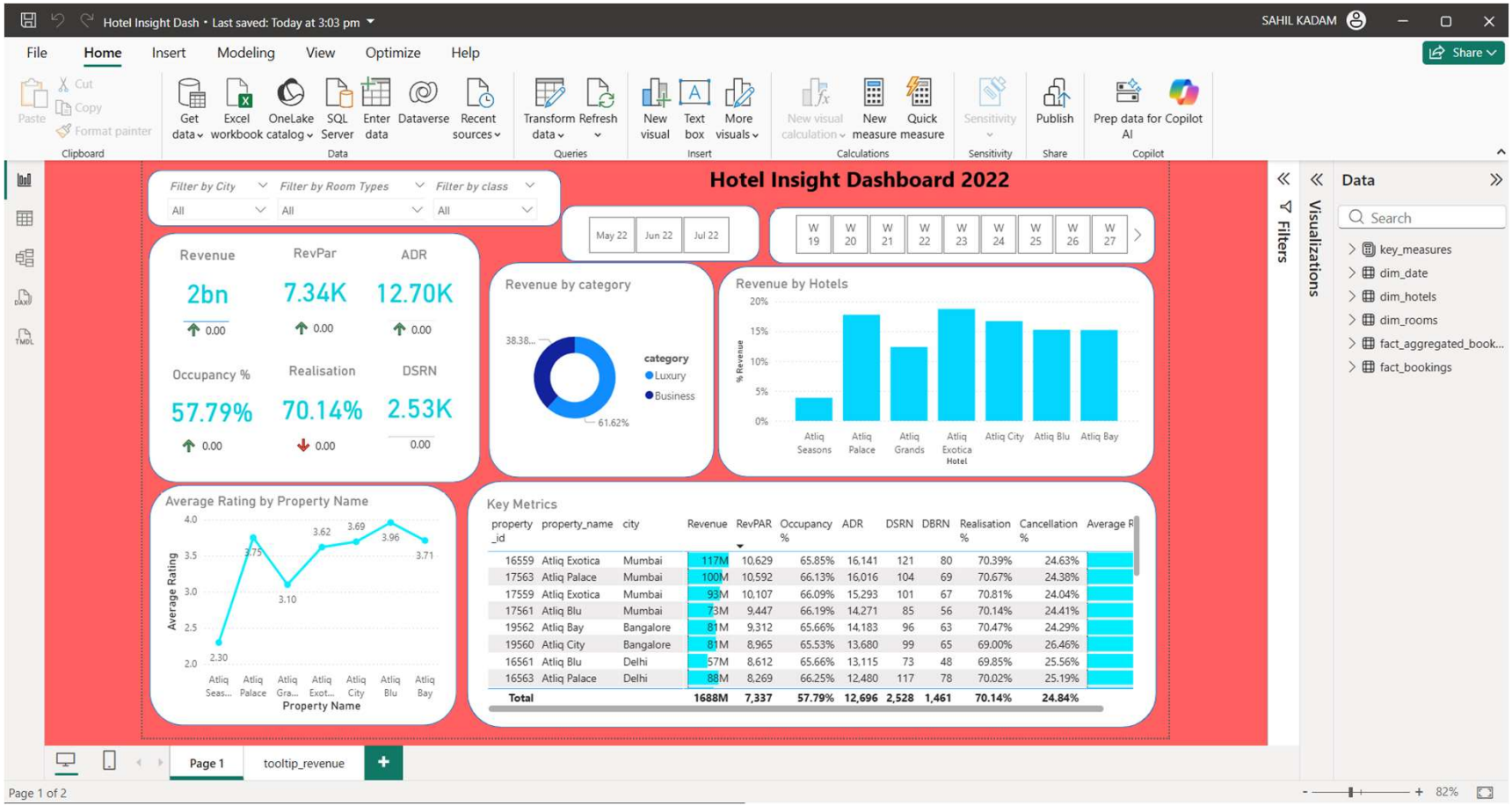
This deployment enables stakeholders to easily monitor hotel performance, analyze trends, and make data-driven business decisions from anywhere.

RESULT:



property_id	property_name	city	Revenue	RevPAR	Occupancy %	ADR	DSRN	DBRN	Realisation %	Cancellation %	Average R
16559	Atliq Exotica	Mumbai	117M	10,629	65.85%	16,141	121	80	70.39%	24.63%	
17563	Atliq Palace	Mumbai	100M	10,592	66.13%	16,016	104	69	70.67%	24.38%	
17559	Atliq Exotica	Mumbai	93M	10,107	66.09%	15,293	101	67	70.81%	24.04%	
17561	Atliq Blu	Mumbai	73M	9,447	66.19%	14,271	85	56	70.14%	24.41%	
19562	Atliq Bay	Bangalore	81M	9,312	65.66%	14,183	96	63	70.47%	24.29%	
19560	Atliq City	Bangalore	81M	8,965	65.53%	13,680	99	65	69.00%	26.46%	
16561	Atliq Blu	Delhi	57M	8,612	65.66%	13,115	73	48	69.85%	25.56%	
16563	Atliq Palace	Delhi	88M	8,269	66.25%	12,480	117	78	70.02%	25.19%	
Total			**1688M**	**7,337**	**57.79%**	**12,696**	**2,528**	**1,461**	**70.14%**	**24.84%**	

RESULT:


Hotel Insight Dash • Last saved: Today at 3:03 pm
SAHIL KADAM
File
Home
Insert
Modeling
View
Optimize
Help
Share
Hotel Insight Dashboard 2022
Filter by City
Filter by Room Types
Filter by class
Revenue
2bn
RevPar
7.34K
ADR
12.70K
Occupancy %
57.79%
Realisation
70.14%
DSRN
2.53K
Revenue by category
Revenue by Hotels
Average Rating by Property Name
Key Metrics
Data
Filters
Visualizations
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tooltip_revenue
Page 1 of 2

CONCLUSION:

- The Hotel Insight Dashboard provides a comprehensive view of hotel performance by consolidating key operational and financial metrics into a single interactive platform. By visualizing important indicators such as Revenue, RevPAR, ADR, Occupancy Rate, Realisation Percentage, and Customer Ratings, the dashboard helps stakeholders quickly understand the performance of different hotels across cities.
- Through interactive filters and clear visualizations, users can easily compare properties, identify high-performing hotels, and detect areas that require improvement. This enables hotel management to make informed, data-driven decisions that can improve operational efficiency, increase occupancy, and enhance customer satisfaction.

Overall, the dashboard demonstrates how business intelligence and data visualization tools can transform raw hotel data into meaningful insights that support strategic planning and better business performance.

FUTURE SCOPE:

The Hotel Insight Dashboard can be further enhanced by integrating additional data sources and advanced analytics to provide deeper business insights. Future improvements may include incorporating real-time booking data, customer feedback analysis, and predictive analytics to forecast occupancy rates and revenue trends.

Machine learning techniques can also be implemented to predict customer demand, optimize pricing strategies, and recommend operational improvements. Integration with hotel management systems and online booking platforms could enable automatic data updates and more accurate performance tracking.

Additionally, the dashboard can be expanded with mobile accessibility, advanced reporting features, and AI-driven insights to help hotel managers make faster and more strategic decisions. These enhancements will further improve decision-making, operational efficiency, and overall hotel business performance.

REFERENCES:

•List and cite relevant sources, data repositories (e.g., the Kaggle dataset you used), and articles that were instrumental in developing the proposed solution.

•Microsoft Power BI Documentation.

•**GitHub Link: https://github.com/24sahil46/Hotel-Insight-Dashboard.git**

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Thank You